

SAN MARCOS, TX, USA

WMO: 722539

Lat: 29.891N

Lon: 97.864W

Elev: 182

StdP: 99.16

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12979

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	-1.0	-12.4	1.3	4.0	-10.1	1.6	4.4	12.7	9.7	11.7	12.0	3.7	0	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	37.8	23.5	37.2	23.5	36.1	23.6	25.8	32.5	25.5	32.2	25.2	31.8	4.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.5	27.9	23.9	19.2	27.9	23.3	18.4	27.7	81.3	32.8	79.2	32.3	77.6	31.9	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.5	8.6	-5.4	39.5	1.7	1.7	-6.6	40.7	-7.6	41.7	-8.5	42.7	-9.7	43.9		
(5)			-6.9	26.7	1.7	0.9	-8.2	27.4	-9.2	27.9	-10.1	28.4	-11.4	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.6	10.5	12.9	16.4	20.7	25.1	28.2	29.0	29.5	26.4	21.3	15.9	11.3		
	DBStd	7.90	5.07	5.69	5.09	4.13	3.18	2.14	1.85	1.74	3.01	4.39	5.01	5.03		
	HDD10.0	149	56	31	9	1	0	0	0	0	1	9	43			
	HDD18.3	889	248	166	98	23	2	0	0	0	1	22	105	224		
	CDD10.0	4029	70	113	208	321	467	545	589	604	491	351	186	84		
	CDD18.3	1726	3	14	38	93	211	295	331	346	242	114	32	8		
	CDH23.3	18789	24	98	266	789	2029	3412	3992	4358	2527	1041	216	39		
(13)	CDH26.7	9232	2	20	57	252	876	1750	2171	2451	1243	373	36	2		
(14)	Wind	WSAvg	4.0	4.3	4.5	4.7	4.8	4.4	3.6	3.4	3.2	3.6	3.8	4.0		
(15) Precipitation	PrecAvg	907	55	58	44	72	126	112	48	62	91	84	81	54		
	PrecMax	1261	208	185	133	206	334	583	230	186	188	315	330	192		
	PrecMin	506	2	7	0	4	7	0	0	1	7	10	0	5		
	PrecStd	219	45	40	30	54	88	108	57	54	53	72	73	49		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	28.9	30.9	33.9	36.2	37.9	38.6	39.0	38.8	33.8	29.4	26.6		
		MCWB	16.7	17.7	19.0	20.6	23.9	23.6	23.8	23.6	22.2	22.9	21.4	18.5		
	2%	DB	23.5	26.3	27.9	31.4	34.1	37.0	37.5	37.8	36.2	32.3	27.6	24.1		
		MCWB	16.1	17.1	18.9	20.8	23.3	23.6	23.5	23.8	23.0	22.7	19.9	17.2		
	5%	DB	22.0	23.8	26.4	29.1	32.8	35.5	36.6	37.2	34.1	30.9	26.1	22.5		
		MCWB	15.8	16.6	18.4	20.2	23.2	24.1	23.5	23.7	23.0	21.9	18.8	16.8		
	10%	DB	19.7	22.1	24.0	27.7	31.8	33.9	35.2	36.1	32.8	28.7	23.9	20.9		
		MCWB	14.9	16.6	17.5	19.6	23.2	23.8	23.7	23.9	22.9	21.3	18.3	16.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.8	20.7	22.1	23.9	26.1	26.8	26.2	26.2	25.8	25.5	22.9	20.8		
		MCDB	23.4	24.2	27.0	29.8	32.8	32.7	33.5	34.3	31.9	30.2	27.1	24.2		
	2%	WB	18.7	19.8	20.9	22.9	25.2	25.9	25.6	25.6	25.4	24.5	21.8	19.7		
		MCDB	21.9	23.4	25.5	28.3	31.7	31.9	32.7	33.2	31.5	29.1	25.6	22.7		
	5%	WB	17.4	18.9	20.1	22.0	24.5	25.5	25.2	25.2	24.9	23.8	20.7	18.7		
		MCDB	20.0	22.0	24.2	26.9	30.3	31.7	32.1	32.5	30.8	28.1	24.4	21.7		
	10%	WB	15.6	17.7	19.1	21.2	23.8	25.0	24.8	24.9	24.3	22.9	19.7	16.9		
		MCDB	18.9	20.6	23.0	26.0	29.2	31.1	31.5	32.1	29.8	26.9	23.3	19.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	12.2	12.2	12.5	11.3	11.4	11.7	12.5	11.8	12.3	12.2	12.5		
		MCDBR	15.4	14.7	13.4	13.7	12.6	12.8	13.5	13.9	13.6	13.1	13.1	14.9		
		MCWBR	10.3	9.0	7.5	6.6	4.8	3.8	3.5	3.8	4.8	6.0	7.7	10.0		
		MCDBR	11.9	10.9	11.1	9.9	10.7	10.6	11.3	12.1	10.9	10.5	10.5	11.7		
	5% WB	MCWBR	9.3	7.6	6.9	5.5	4.7	4.0	3.8	3.7	4.4	5.7	6.7	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.337	0.337	0.363	0.402	0.449	0.430	0.443	0.429	0.415	0.375	0.362	0.320		
	taud		2.466	2.480	2.418	2.325	2.240	2.344	2.340	2.363	2.358	2.451	2.425	2.532		
	Ebn at Noon		896	930	923	893	848	859	846	857	859	876	859	900		
	Edh at Noon		93	100	113	129	141	127	127	123	119	102	96	83		
(44) All-Sky Solar Radiation	RadAvg		3.00	3.49	4.48	5.36	5.78	6.58	6.62	6.28	5.18	4.30	3.23	2.66		
	RadStd		0.33	0.62	0.42	0.49	0.51	0.42	0.55	0.34	0.49	0.57	0.37	0.34		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+82	

Nomenclature: See separate page